

Baselines for Small Defect Detection

Dataset Overview

Seven public industrial defect datasets were combined into a single benchmark of 12,670 images, each with both a bounding-box label and a binary segmentation mask. Every instance is bucketed by defect size (small / medium / large, relative to image area) for size-stratified evaluation.

Dataset	Surface / Domain	Small	Medium	Large	Total
DAGM	Synthetic textured industrial surfaces	593	1,095	412	2,100
GC10-DET	Steel / metal strip surfaces	322	622	1,348	2,292
KolektorSDD2	Electrical commutator production surfaces	77	177	102	356
MPDD	Painted metal / product surfaces	123	92	67	282
MTD	Magnetic tile surfaces	188	57	143	388
Severstal	Steel surface defects	1,618	3,656	1,821	7,095
VisA	Mixed industrial object / surface categories	114	42	1	157
Total		3,035	5,741	3,894	12,670

Split methodology: a fixed 70/15/15 train/validation/test split, stratified jointly by dataset and size bucket (i.e. by dataset_sizebucket, e.g. "DAGM_small") so every dataset and every size class is proportionally represented in each split. Fixed random seed (42), identical split reused across every baseline for direct comparability.

Split	Count
Train	8,858
Validation	1,892
Test (overall)	1,920
Test — small	462
Test — medium	868
Test — large	590

Detection Baselines

Model	Batch	Epochs	mAP50	mAP50-95	Precision	Recall	mAP50 Small	mAP50 Medium	mAP50 Large	Recall Small	Recall Medium	Recall Large	Infer. (ms)	FP / Image
YOLOv8n	8	100	0.690	0.371	0.710	0.631	0.617	0.741	0.674	0.591	0.666	0.614	2.40	0.790
YOLOv9t	16	100	0.237	0.113	0.337	0.429	0.105	0.262	0.201	0.247	0.450	0.477	18.378	1.830
YOLOv10n	8	100	0.668	0.357	0.696	0.599	0.578	0.731	0.641	0.542	0.639	0.571	2.561	0.694
RF-DETR	16	42	0.646	0.351	0.547	0.738	0.314	0.588	0.645	0.557	0.750	0.800	31.71	1.328
D-FINE-S	8	50	0.645	0.348	0.711	0.582	0.596	0.679	0.627	0.543	0.613	0.562	—	0.525
YOLO-World	16	85	0.226	0.118	0.230	0.392	0.108	0.207	0.160	0.315	0.389	0.427	24.072	2.705
YOLOv11s	16	90	0.433	0.204	0.447	0.577	0.439	0.547	0.513	0.577	0.012	0.053	20.227	0.960

Segmentation Baselines

Model	Batch	Epochs	Precision	Recall	Recall Small	Recall Medium	Recall Large	Infer. (ms)	FP / Image	Dice	IoU
SAM	16	45	0.993	0.993	0.963	0.995	1.000	321.94	0.015		
SAM2	16	50	0.979	0.979	0.964	0.988	1.000	109.27	0.044		
SegFormer-B0	8	21	0.786	0.729	0.512	0.673	0.750	22.986	5062.95	0.757	0.609
SegNeXt-T	8	41	0.766	0.753	0.565	0.737	0.762	23.490	5870.70	0.759	0.612
GCNet-S	8	29	0.712	0.692	0.488	0.667	0.704	14.522	7137.88	0.702	0.541
PIDNet-S	8	31	0.739	0.586	0.414	0.607	0.586	6.836	5290.07	0.654	0.486
Mask2Former	4	39	0.779	0.784	0.714	0.770	0.789	63.877	5550.21	0.781	0.641

Code Repository

<https://github.com/ananyashriram10/small-defect-detection>